# Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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# Agenda

**Introduction**

Language Overview

Syntax

Metatheory

Conclusion

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# Background

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    - Most useful as *thing*
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# Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

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- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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# Existing Solutions

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- **Resource Description Language (RDF)**
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - — No built-in logic engine
- **Labeled Property Graphs (LPGs)**
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - — Not standardized
- **Cyc**
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - — Hard-coded, proprietary rules

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# Proposed Solution

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- **Welkin**: an information language

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# General Requirements

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# General Requirements

1. Express any computer program

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   - Established notion: **Turing machines**[1]
2. Enable user-defined *meaning*

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- Achieve both through **representations**

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- Centered around *representing* things
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- Track *faithfulness* of representations through **truth**

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- Representations are units
- Each unit has its own **context**

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# Information

- Representations

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## Information

- Representations ~ **Truth**
- Information

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## Information

- Representations ~ **Truth**
- Information ~ **Proofs**
-

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## Rules

- **Internal Transitivity:**
- **Contextual Lifting:**

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## Rules

- **Internal Transitivity:**
- **Contextual Lifting:**
- **Lattice Laws:**
  - Units as sets

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# Turing Completeness

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## Turing Completeness

- Part of Universality
- Interpretation: $a - c \to b$ iff $\varphi_c(\langle \varphi_a \rangle) = \langle \varphi_b \rangle$

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## Encoding

- Characters written in `type-writer font`[1]
- Define US-ASCII: Printable + `EOF` (code: 58)

[1] Font used: Intel One Mono (https://github.com/intel/intel-one-mono), licensed under OFL 1.1 (https://github.com/intel/intel-one-mono/blob/main/OFL.txt).

# Invertible Syntax Descriptions

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# The Welkin Grammar

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# Outline of Unambiguity

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